Production Planning Optimization App

📌 Background

Manufacturing and supply chain businesses often face the challenge of deciding how much of each product to produce when resources such as raw materials and machine hours are limited. Producing too much of one product can exhaust critical resources, while underproducing another can reduce overall profitability.

Optimization methods, particularly **Linear Programming (LP)**, provide a mathematical way to balance these trade-offs and maximize profits under resource constraints.

Problem Statement

Given:

- A set of **products** (each with its selling price and unit cost).
- A set of **resources** (raw materials, machine hours, etc., each with limited availability and cost).
- A **mapping** between products and resources (how much of each resource is consumed per unit of product).

Question:
How can we determine the **optimal production plan** that **maximizes profit** while respecting all resource constraints?

🛠️ Approach

This project applies **Linear Programming (LP)** using the **PuLP** solver in Python.

Steps:

1. **Data Inputs** (CSV files in `/data`):
   - `products.csv`: List of products with price and unit cost.
   - `resources.csv`: Available resources, their capacity, and cost per unit.
   - `product_resources.csv`: Defines how much of each resource is required to produce one unit of each product.
2. **Model Setup**:
   - Decision variables: how many units of each product to produce.
   - Objective: Maximize total profit = (price – cost – resource usage cost).
   - Constraints: Do not exceed available resource capacity.
3. **Solution**:
   - Uses PuLP to solve the LP.
   - Returns production quantities, total profit, and resource utilization.
4. **Visualization**:

- Built with **Streamlit** for interactive results.
- Displays optimal product mix, resource usage, and profit.

📂 Folder Structure

```
production-planning/
│
├── app.py              # Streamlit app
├── model.py            # Optimization model (PuLP)
├── data/
│   ├── products.csv
│   ├── resources.csv
│   └── product_resources.csv
└── README.md           # Documentation
```

The solver will calculate the **optimal production plan** to maximize profit given these constraints.

🚀 Future Improvements

- Add **multi-period planning** (weekly/monthly horizon).
- Include **demand constraints** (minimum/maximum product demand).
- Add **inventory tracking** across periods.
- Enable **sensitivity analysis** (e.g., "What if we had more machine hours?").
- Export results to **Excel or PDF** for reporting.

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